

Ex 1.

1) Identify the entities.

Case 1: Hospital Management System

In a hospital management system, various entities interact to ensure smooth operations and quality patient care. Patients are at the core of the system, each identified by their unique patient ID. Patient records contain essential personal details such as name, date of birth, gender, contact information, and medical history. Doctors play a crucial role in providing medical services to patients. They are identified by their physician ID and possess attributes like name, specialty, contact details, and qualifications. Doctors are associated with specific medical departments within the hospital, each department representing a distinct area of medical expertise.

Appointments facilitate the scheduling of patient consultations with doctors. An appointment record includes the appointment ID, date and time, patient ID, doctor ID, and the purpose of the visit. Patients can have multiple appointments with different doctors for various medical concerns. The hospital's medical departments are integral to organizing patient care efficiently. Each department is characterized by a department ID, name, and description, representing specialties such as cardiology, pediatrics, orthopedics, etc. Doctors are assigned to specific departments based on their medical expertise and patient needs.

Patient	Doctor	Department	Appointment
> Patient ID P.K	> Physician ID P.K	> Department ID P.K	> Appointment ID P.K
> Name	> Name	> Name	> Date - Time
> DOB	> Specialty	> Description	> Purpose
> Gender	> Contact Details		> Patient ID FK
> Contact Info	> Qualifications		> Physician ID FK
> Medical History	> Department FK		

3. Relationships

4. Cardinalities

* Department has Doctor.

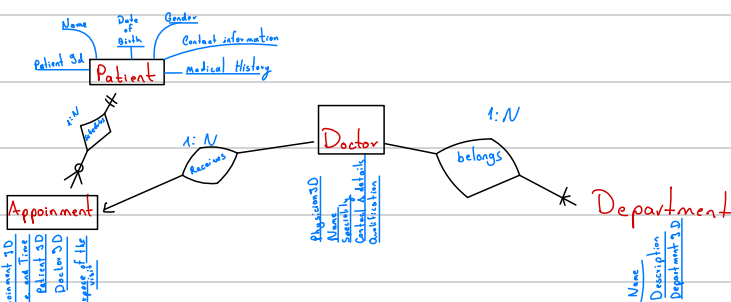
* Department → Doctor (1:N)

* Patient books appointment.

* Patient → appointment 1:N

* Doctor attends appointment.

* Doctor ~ appointment 1:N



Case 2: Online Bookstore Management System

In an online bookstore management system, various entities collaborate to provide customers with access to a wide range of books. Customers are the focal point of the system, each having a unique customer ID. Customer profiles contain personal information such as name, email address, phone number, and shipping address. Books are the primary products offered by the bookstore, each identified by a unique ISBN (International Standard Book Number). Book records include attributes like title, author(s), genre, publication date, and price. Multiple copies of the same book can exist in the inventory.

Orders represent customer requests to purchase books from the bookstore. Each order is associated with a unique order ID and includes details such as the customer ID, book ISBN, quantity, and order date. Customers can place multiple orders for different books. Publishers play a significant role in supplying books to the bookstore. Each publisher is identified by a publisher ID and has attributes such as name, location, and contact information. Publishers may have partnerships with various authors and produce books across different genres.

Identify the entities

Customer	Book	Publisher	Order	Inventory
> Customer ID P.K	> ISBN P.K	> Publisher ID P.K	> Order ID P.K	Null
> Name	> Title	> Name	> Order Date	
> Mail	> Authors	> Location	> Quantity	
> Phone	> Genre	> Contact Info	> Customer ID FK	
> Shipping address	> Publication Date			
	> Price			
	> Publisher ID FK			

* Customer places Order

* Customer places Order 1:N

* Order includes Book

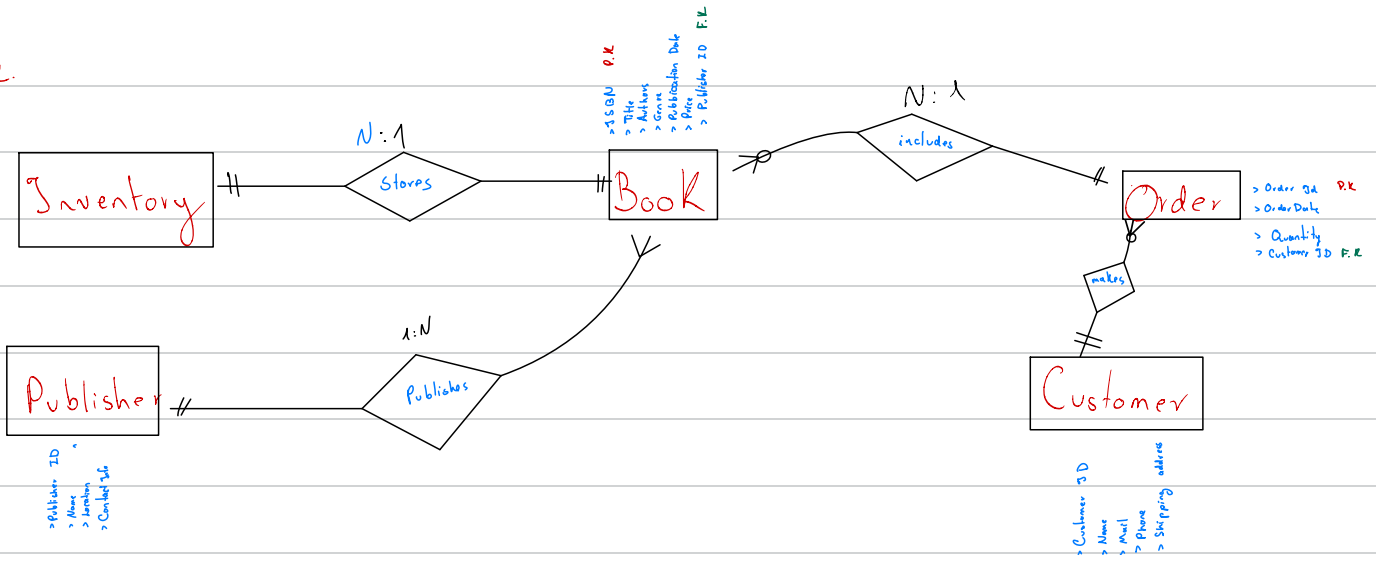
* Order includes Book 1:N

* Book is stocked in inventory

* Book is stocked in inventory N:1

* Order Book N:1

2.



3.

Case 3: Person-Children Relationship

Consider the following PERSON-HAS CHILDREN-PERSON relationship. A person can have many children or none. A person is always the child of another person. The attributes of the person are identity card, name, address and telephone number. Transform it to the relational model.

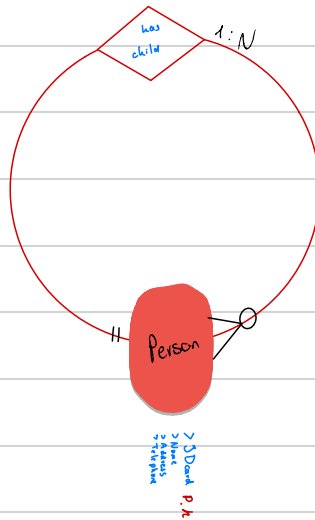
1. Entity

2. Attributes

- > ID card P.K
- > Name
- > Address
- > Telephone

3. Person has child
> A person can have many children or none

1:1



4.

A growing entrepreneurship operates in multiple regions and needs a database to manage its operations and staff. The company is led by a Chief, who supervises several Regional Managers. Each Regional Manager oversees multiple Departments (such as HR, IT, and Warehouse), ensuring the teams meet business goals.

X

The company employs many Employees, each assigned to exactly one Department. Employees can be assigned to multiple Projects, and each project may involve many employees working together across departments. This requires tracking who works on what project and in what capacity.

The business also sells Products that are handled and managed through operational departments (most commonly Warehouse). Product stock is tracked in Inventory, allowing the company to record and monitor available quantities and stock updates over time. Finally, each employee receives periodic payments tracked in Payroll, storing payment history for accounting and HR compliance.

1. Entities

Chief	R. Manager	Department	Employee	Project	Product	Inventory	Payroll
> Chief ID P.K. > Name > Email > Phone	> M. ID P.K. > Name > Region > Mail > Phone	> D. ID P.K. > Name > Description	> Employee ID P.K. > Name > Mail > Phone > Hire Date	> P. ID P.K. > Name > Start Date > End Date > Budget	> P. ID P.K. > Name > Category > Price	> Inventory ID P.K. > P. ID F.K. > Quantity > Update	> P. Roll ID P.K. > Employee ID F.K. > Pay Date > Amount > Method

3. Relations

4. Cardinality

- * Chief Manages R.M. 1:N
- * R.M. oversees Department 1:N
- * Department has Employees 1:N
- * Employee works on projects 1:N
- * Products stored in Inventory 1:N
- * Department handles product 1:N
- * Employee receives payroll 1:N

